

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 03

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

CPU Registers

Topic

Types of Architecture

Topic

Program Counter

Topic

Instruction Register

Topic

Stack Pointer

Topics to be Covered



Topic

Memory Addressing

Topic

Memory Access

Topic

Architecture Type (Based on Size of Input)

H.W.

$$(a+b) * (c+d)$$

AC-based arch:-

$$AC \leftarrow a$$

$$AC \leftarrow AC + b$$

$$R1 \leftarrow AC$$

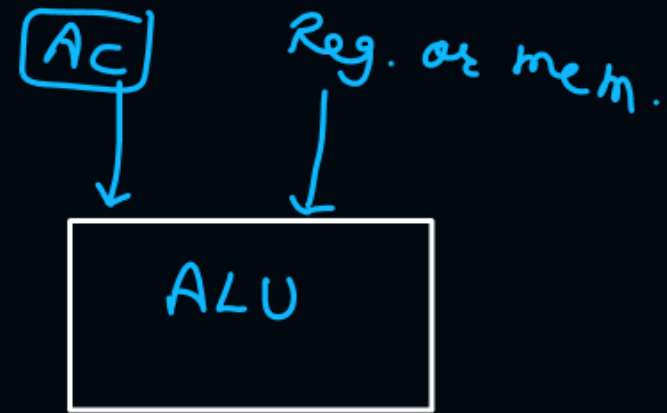
$$AC \leftarrow c$$

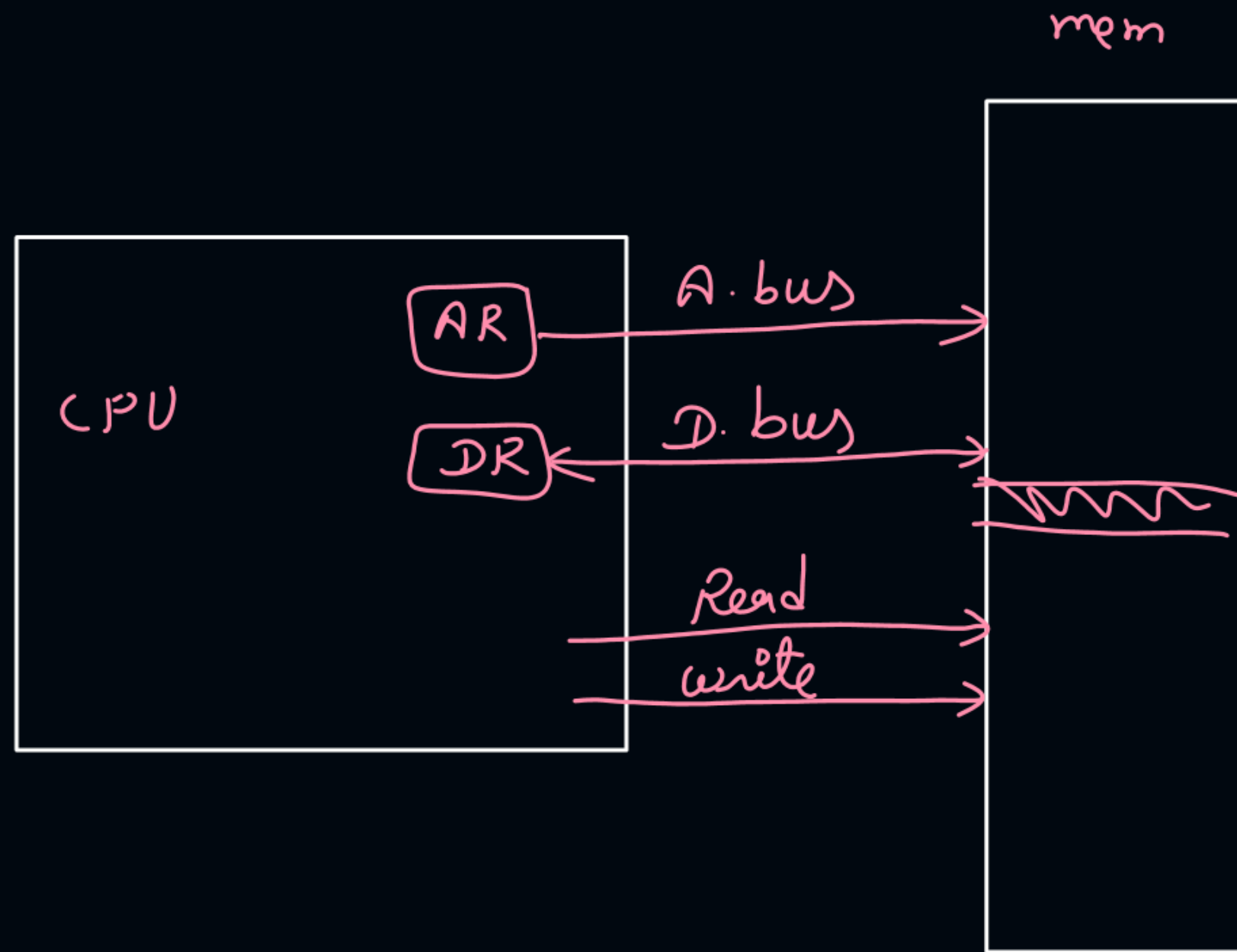
$$AC \leftarrow AC + d$$

$$R2 \leftarrow AC$$

$$AC \leftarrow R1$$

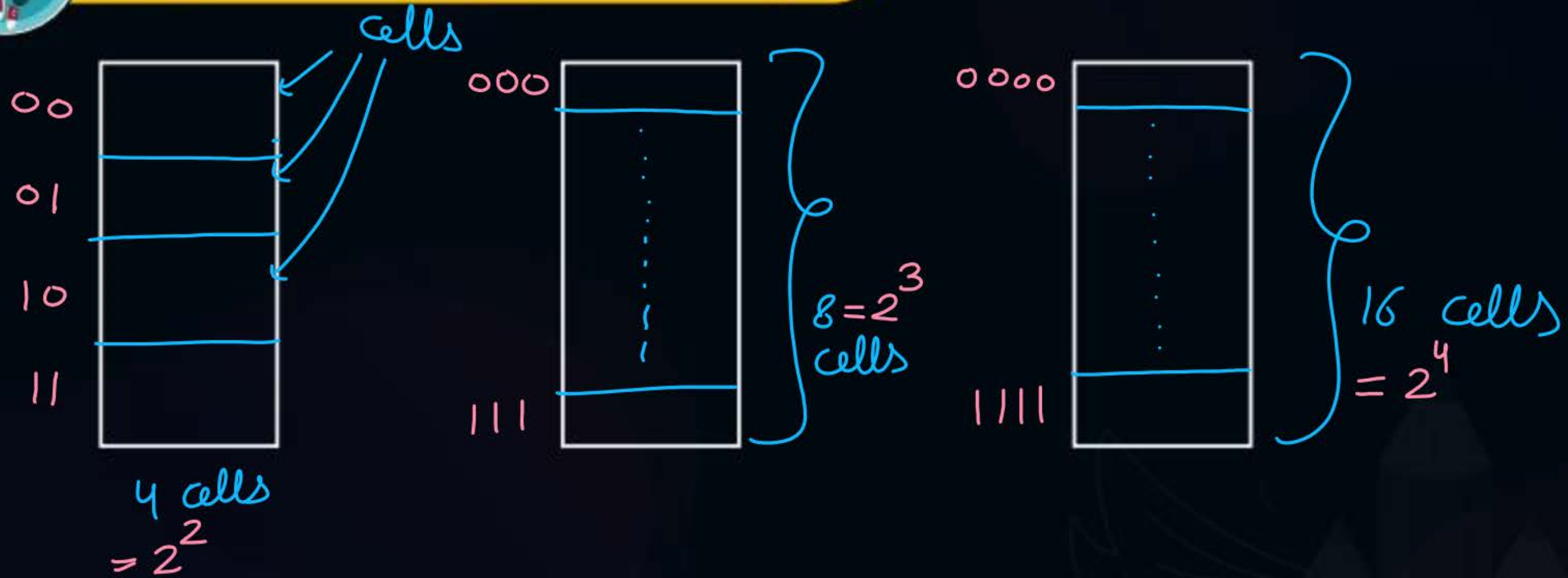
$$AC \leftarrow AC * R2$$







Topic : Memory Addressing





Topic : Memory Addressing

no. of cells	$4 = 2^2$	$8 = 2^3$	$16 = 2^4$	2^n	x
address length	2 bits	3 bits	4 bits	n bits	$\log_2 x$ bits

$$\log_2 2^n = n \text{ bits}$$

#Q. Number of cells in memory = 2^8

Address size for memory = 8 bits?

#Q. Number of cells in memory = 512 = 2^9
Address size for memory = 9 bits?

Ques) no. of cells = 4K
add. size = 12 bits?

Solⁿ

$$4 * 2^{10} = 2^2 * 2^{10} = 2^{12} \Rightarrow \text{add.} = 12 \text{ bits}$$

#Q. Number of cells in memory = 32M = $2^5 * 2^{20} = 2^{25}$
Address size for memory = 25 bits?



#Q. Number of address bits to access memory = 8 bits

Number of cells in memory = $\frac{2^8}{8} = 256$?



#Q. Number of address bits to access memory = 12 bits

Number of cells in memory = 4K?

$$\text{no. of cells} = 2^{12} = 2^2 * 2^{10} = 4K$$

#Q. Number of address bits to access memory = 27 bits

Number of cells in memory = 128 M?

$$\text{no. of cells} = 2^{27} = 2^7 * 2^{20} = 128 \text{ M}$$



Topic : Memory Types

- default
1. Byte Addressable $\Rightarrow$ on each cell in memory one byte is stored.
 2. Word Addressable $\Rightarrow$ on each cell in memory one word is stored.

Word Addressable

1 word
1 word
1 word
⋮
1 word

1 word
can be 1 byte
or 2 bytes
or 4 bytes
or 8 bytes ;

Byte Addressable

1 byte
1 byte
1 byte
1 byte
1 byte
⋮

used in
our systems



Topic : Storing Contents in Memory

for example:-

char x; // each char $\Rightarrow$ 1 byte
int y; // each int $\Rightarrow$ 2 bytes
float z; // each float $\Rightarrow$ 4 bytes

byte addressable

200	x	} 1 byte
500	y	} 2 bytes
501		
600	z	} 4 bytes
603		

word addressable
(1 word = 4 bytes)

200	x	
500	y	
600	z	



Topic : Storing Contents in Memory

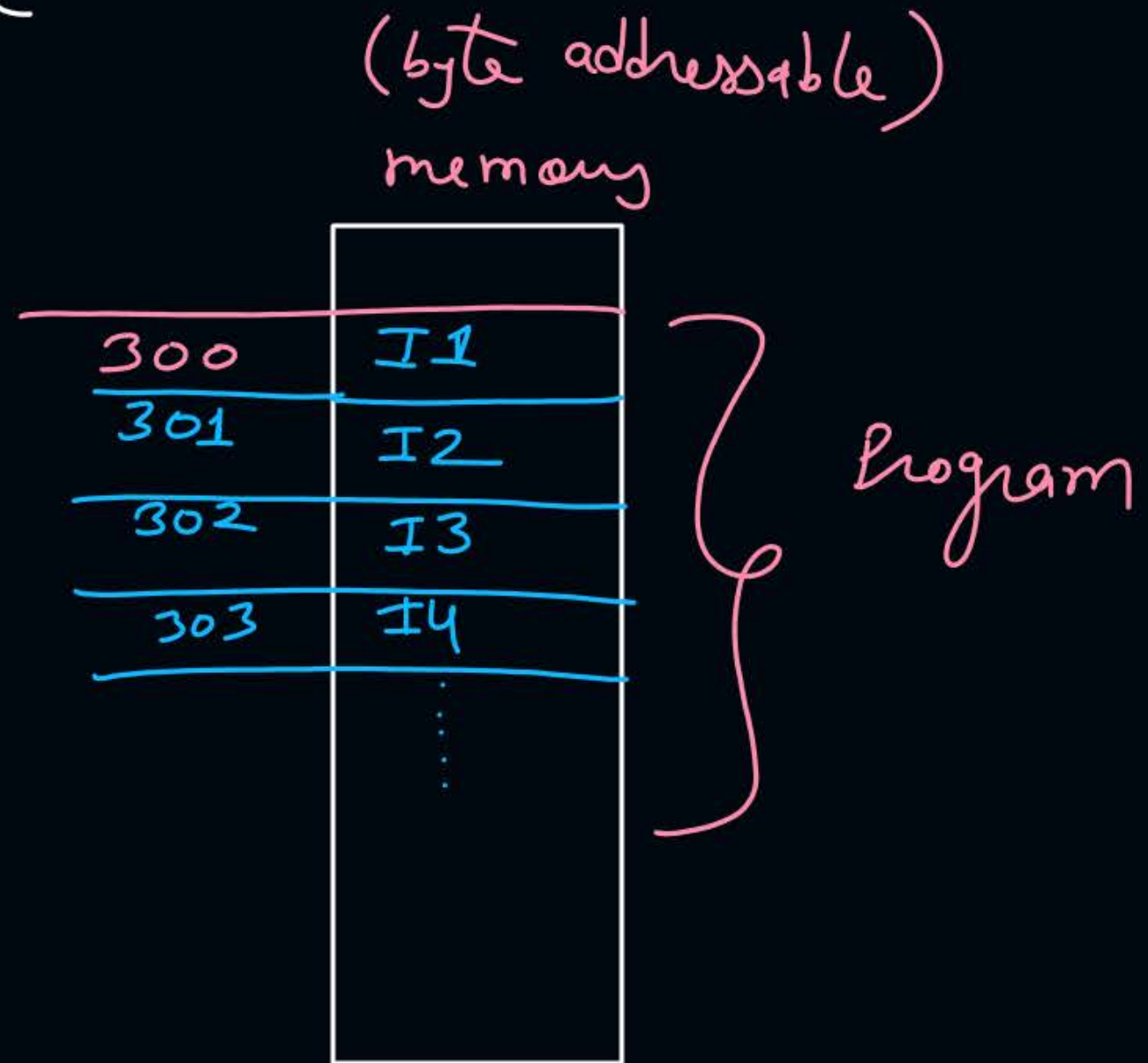
If any content is stored on more than one addresses.
↓
it is represented by it's first add.

ex:- if int y stored on addresses 500 and 501.
↓
int y stored from 500

storing instructions in memory:-

assume each instruction $\Rightarrow$ 1 byte

when instruction I3 is in
CPU for execution then
the value of PC = 303 ?



#Q. A CPU has 4 bytes instructions. A program (Instructions I_1 to I_{200}) starts at address 200 (in decimal). Memory is byte addressable. Find the address of following instructions:

1. $I_1 \Rightarrow 200$

2. $I_5 \Rightarrow 216$

3. $I_{120} \Rightarrow 200 + (120 - 1) * 4$
 $= \underline{\underline{676}} \text{ Ans.}$

200	I_1
204	I_2
208	I_3
212	I_4
216	I_5
	$\vdots$

#Q. A CPU has 4 bytes instructions. A program (Instructions I_1 to I_{200}) starts at address 500 (in decimal). Memory is byte addressable. What should be the PC value when instruction I_6 will be executing in CPU?

$$\begin{aligned} \text{PC value} &= \text{add. of } I_7 = 500 + (7-1) * 4 \\ &= \underline{\underline{524}} \text{ Ans.} \end{aligned}$$

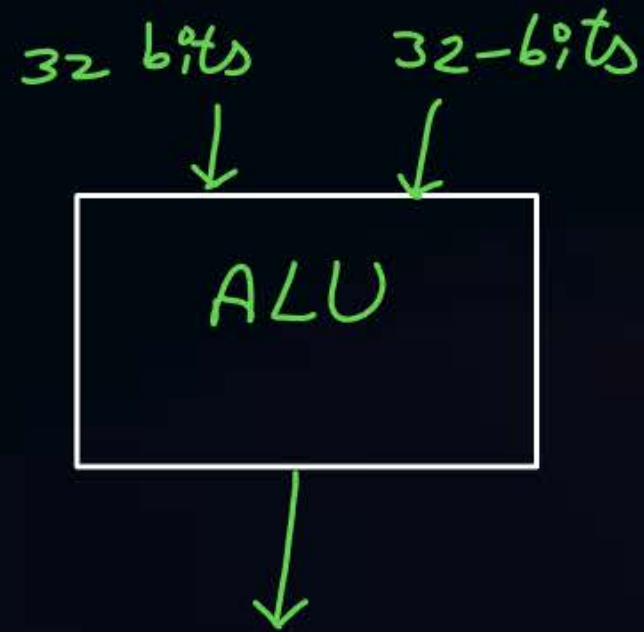
#Q. A CPU has 4 bytes instructions. A program (Instructions I_1 to I_{200}) starts at address 500 (in decimal). Memory is byte addressable. What should be the PC value when instruction i will be executing in CPU?

$$\begin{aligned} PC &= \text{add. of } (i+1) = 500 + (i+1-1) \times 4 \\ &= 500 + 4i \end{aligned}$$



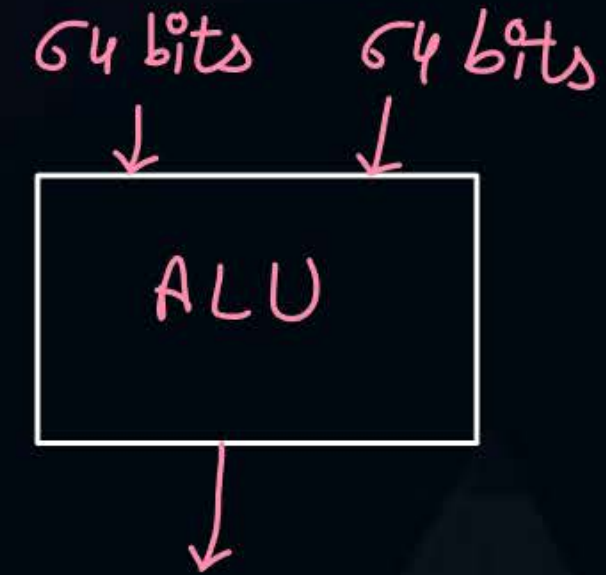
Topic : Architecture Type (Based on Size of Input)

32-bits architecture



CPU word size = 32 bits

64-bits architecture



CPU word size = 64 bits



Topic : Micro Operation

→ smallest operation a CPU can perform at a time



- The operations executed on values stored in registers
- Symbolic Notation to describe the micro-ops: **Register Transfer Language (RTL)**

ex:-

content of GPR R1 is copied to
GPR R2.

$R2 \leftarrow R1$ or $R1 \rightarrow R2$

$R1 = 5$
 $R2 = 5$



Topic : Micro Operation

- Register Transfer: $R2 \leftarrow R1$ or $R1 \rightarrow R2$
- Comma: $IR \leftarrow DR, PC \leftarrow PC + 1$ $\Leftarrow$ these 2 micro-operations can be performed simultaneously
- Memory Transfer:

Read :-

$CPU \leftarrow \text{memory}$

$DR \leftarrow M[\text{address}]$

$DR \leftarrow M[1000]$

or
 $DR \leftarrow M[AR]$

$AR = 200$

$DR \leftarrow M[AR]$

$DR = 15$

ex:- mem.



200

write :-

$M[Address] \leftarrow DR$



Topic : Example

$$R1 = \cancel{15} \overset{20}{-25}$$

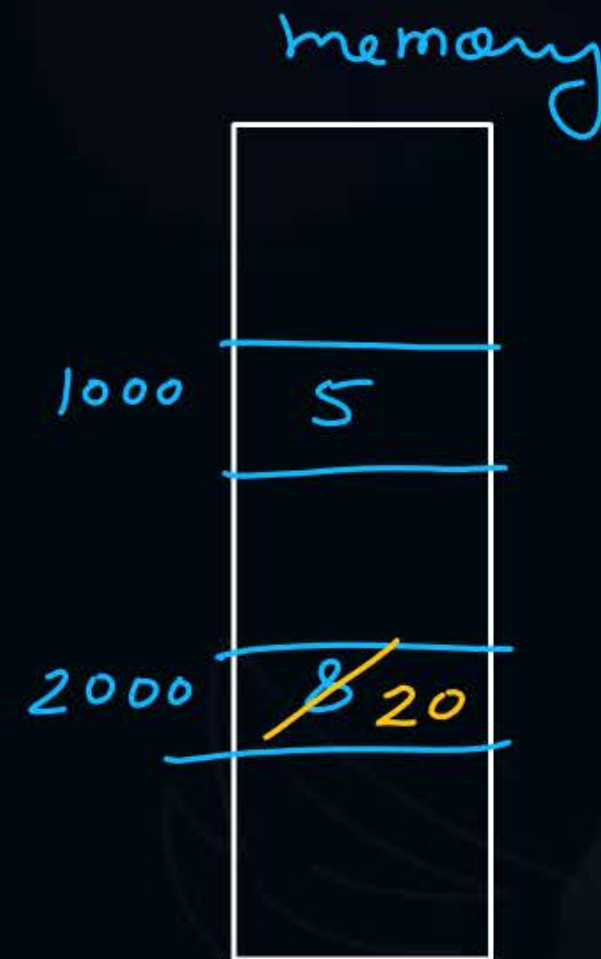
$$R2 = \cancel{10} \underset{5}{5}$$

$$R1 \leftarrow R1 + R2$$

$$R2 \leftarrow m[1000]$$

$$R1 \leftarrow R1 - R2$$

$$m[2000] \leftarrow R1$$



#Q. Consider the following program segment. Here R1 and R2 are the general purpose register. Assume that the content of memory location 2000 is 20. All numbers are in decimal. After the execution of this program the value of memory location 2000 is?

Instructions	Operations
MOV R1, #12	$R1 \leftarrow \#12$
MOV R2, (2000)	$R2 \leftarrow M[2000]$
SUB R2, R1	$R2 \leftarrow R2 - R1$
MOV (2000), R2	$M[2000] \leftarrow R2$
HALT	Stop

#Q. Consider the following program segment. Here R1 and R2 are the general purpose register. Assume that the content of memory location 3000 is 23. All numbers are in decimal. After the execution of this program the value of memory location 3000 is?

Instructions	Operations
MOV R1, #11	$R1 \leftarrow \#11$
MOV R2, (3000)	$R2 \leftarrow M[3000]$
ADD R2, R1	$R2 \leftarrow R2 + R1$
ADD R1, R2	$R1 \leftarrow R1 + R2$
MOV (3000), R1	$M[3000] \leftarrow R1$
HALT	Stop



2 mins Summary



Topic

Memory Addressing

Topic

Memory Access

Topic

Architecture Type (Based on Size of Input)



Happy Learning

THANK - YOU

